



Use Owl components on the portal and website

In this article, you will learn how you can leverage Owl components on the portal and website.

Overview

To use Owl components on the website or portal, you will need to do a few things:

- Create your Owl component and register it in the `public_components` registry
- Add that component to the `web.assets_frontend` bundle
- Add an `<owl-component>` tag to a website or portal page to use the component

1. Creating the Owl component

To keep things simple, let's start with a very straightforward component that just renders "Hello, World!". This will let us know at a glance if our setup is working.

First, create the template in `/your_module/static/src/portal_component/your_component.xml`.

```
<?xml version="1.0" encoding="UTF-8"?>
<templates xml:space="preserve">
  <t t-name="your_module.YourComponent">
    Hello, World!
  </t>
</templates>
```

Then create the JavaScript file for that component in

`/your_module/static/src/portal_component/your_component.js`, and add it to the `public_components` registry:

```
import { Component } from "@odoo/owl";
import { registry } from "@web/core/registry"

export class YourComponent extends Component {
  static template = "your_module.YourComponent";
  static props = {};
}
```

See also

Owl components reference.

2. Adding your component to the `web.assets_frontend` bundle

The `web.assets_frontend` bundle is the assets bundle that is used by the portal and website, you'll want to add your component's code to that bundle so that the public components service can find your component and mount it. In your module's manifest, in the `assets` section, add an entry for `web.assets_frontend` and add your component's files:

```
{
    # ...
    'assets': {
        'web.assets_frontend': [
            'your_module/static/src/portal_component/**/*.js',
        ],
    }
}
```

See also

Module manifest reference.

3. Adding an `<owl-component>` tag to a page

Now we need add an `<owl-component>` tag that will serve as the target for the component to be mounted. For the sake of this example, we'll add it directly to the portal's home page with an xpath in `/your_module/views/templates.xml`.

```
<?xml version="1.0" encoding="utf-8"?>
<odoo>
    <template id="your_module.portal_my_home" inherit_id="portal.portal_my_home">
        <xpath expr="//*[hasclass('o_portal_my_home')]" position="before">
            <owl-component name="your_module.YourComponent" />
        </xpath>
    </template>
</odoo>
```

Don't forget to add this file to the data section of your assets bundle:

```
{
    # ...
    'data': [
        'views/templates.xml',
    ]
}
```

And that's it! If you open the home page of the portal you should see the message "Hello, World!" at the top of the page.

Points of caution

Owl components are rendered entirely in JavaScript by the browser. This can cause some issues:

- Layout shift
- Poorer indexing by search engines

For these reasons, you should only use Owl components on the portal and website for specific use cases described below.

Layout shift

When a page initially renders content, and that content subsequently moves ("shifts") within the page, this is referred to as a layout shift. When using Owl components in the portal or website, all of the HTML that surrounds the Owl component is rendered by the server and is the first thing that will be displayed to the user. When JavaScript starts running, Owl will mount your component, which will likely cause the surrounding elements to move around on the page. This can cause poor user experience: the user sees an element on the page that was initially rendered and with which they want to interact, so they move their cursor or finger above that element. Just as they're about to click, the Owl component is mounted, and the element they want to interact with is moved. They click and interact with the Owl app instead.

This can be a frustrating experience, so you should be careful when designing your page that the Owl component will not move elements around. This can be achieved in various ways, e.g.

See also

[Cumulative Layout Shift on web.dev](#) 

Poorer indexing by search engines

When search engines build their index of the content of the web, they use web crawlers to find pages and analyze their content to show these pages in their search results. While modern search engines are generally capable of executing JavaScript code and should generally be able to view and index content rendered in JavaScript, they may not index the content as fast and penalize the page in search results.

Because most search engines do not reveal the exact way that they crawl and index web pages, it's not always easy to know the extent of the impact that client-side rendering can have on your search engine scores. While it's unlikely to make or break your SEO strategy, you should still only use Owl components when they are adding real value over server-side rendering.

When to use Owl components on the portal and website

As explained in the previous sections, using Owl component can slightly degrade user experience if you're not careful and may also hinder your SEO. So when should you choose to use Owl components in these places? Here are some general guidelines.

When you don't care about SEO

If a page cannot be indexed by search engines because it's not available to the public, e.g. anything in the user portal, SEO performance is not a concern, as web crawlers cannot access these pages anyway. There are also some things that you do not want or care about indexing, e.g. if you want to have a page where the user can choose a date and time for an appointment, you probably don't want search engines to index the dates on which an appointment is possible at a specific moment in time.

When you need strong interactivity

The decision to use Owl is a trade-off between the previously mentioned disadvantages and the effort that Owl saves you by making it easy to build richly interactive user experiences. The main reason to use Owl is when you want to build an interface that can react in real time to



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